AETHERFORGE

Version 0.34 · Beta

Sail the luminiferous aether in the year 1889. Trade, fight and explore among the worlds, and settle the score with the villain who ruined your family.

CAPTAIN'S MANUAL



"Being a captain is a dangerous job in an unforgiving aether. Being a careless one just makes it the first mate's problem."

Before you launch

AETHERFORGE IS IN BETA

This game is still in active development, and things can change quickly: new features, balance changes and fixes arrive often. Update regularly so you're always playing the latest version. Your saved voyage carries over when you update. If something looks wrong, it may already be fixed in a newer version.

> *"Being a captain is a dangerous job in an unforgiving aether. Being a careless one just makes it the first mate's problem."*
>
> WORDS TO SAIL BY

PLAYING AND UPDATING

- Play in any modern web browser at bugabooman-lgtm.github.io/aetherforge.
- **Get the latest version:** Aetherforge.html always opens the newest release. Save it to keep a copy (see *Playing offline* below).
- **Get the latest manual:** Aetherforge-Manual.pdf is always the newest edition of this manual.
- To check for a new version, choose **Saves, Backups & Updates** on the title screen (or **Backups and updates** in the pause menu) and press **Check for update**.
- Your voyage saves automatically in your browser. Save a backup file before clearing your browser's data (see *Saving* near the end of this manual).
- **Network play** with friends works only on the website above, not in an offline copy.
- Found a bug, or have an idea? Write to GadgetsAndGizmotos@gmail.com.

PLAYING OFFLINE

The whole game is a single file, so once it's saved it runs without an internet connection.

1. Open bugabooman-lgtm.github.io/aetherforge/Aetherforge.html.
2. Save the page with Ctrl+S (Cmd+S on a Mac), as a complete web page or HTML only.
3. Keep the saved **Aetherforge.html** anywhere on your computer and open it to play.

- **Saves stay with where you play.** Your browser keeps the offline file's voyages separate from the website's. To move a voyage between them, use **Save backup file** and **Load backup file** under *Saves, Backups & Updates*.
- **Updating an offline copy:** when you're online, **Check for update** still works. If a newer version is out, **Download** saves the new file with your voyage already inside it. Replace your old file with it.
- Offline, the game uses your computer's own fonts, so text looks slightly different. Offline play works best on a computer; phones make saving web pages awkward.

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About Aetherforge

Aetherforge is an open-world adventure in a solar system that never was: part trading voyage, part privateer's cruise, part Victorian scientific romance. You command a small ether flyer in the year 1889 and make your fortune, and your name, among the worlds.

THE INSPIRATION

In 1990 a computer game called *Space: 1889* took players into the Victorian future imagined by Jules Verne and H. G. Wells, where ships sail the aether to Mars and Venus. A few years earlier, *Sid Meier's Pirates!* had given players an open sea of short adventures: trade, sea battles, sword fights and a family to find. And *Starflight* had captured the joy of setting out into the unknown.

Aetherforge brings the three together. It stays inside our own solar system and keeps to what the Victorians believed about it. Every wonder in the game comes from a real theory of the day that later proved wrong: the luminiferous aether and its wind, apergy, caloric, the canals of Mars, the swamps of Venus and the lost planet Phaeton.

HOW IT WAS MADE

Aetherforge is a sister game to Starforge, and like Starforge it was designed one idea at a time and built in conversation with Claude, an AI from Anthropic. The species art comes from Starforge, restyled in brass and sepia. It runs entirely in a web browser as a single file.

THE STORY SO FAR

Your father was a quiet astronomer who studied the ancient relics found on other worlds. Then the evidence of a theft was laid at his door, and your family was ruined. You have a battered packet sloop, thirty hands and a name to clear. Somewhere out in the aether, the one who wronged you is waiting.

Welcome

WELCOME ABOARD, CAPTAIN

It is 1889, and the luminiferous aether is real. Ether flyers driven by steam and apergy sail between the worlds, and the great powers race to claim Mars, Venus and Mercury. You begin with a battered packet sloop, thirty hands and a debt to settle with the villain who ruined your family.

- **Trade.** Buy where goods are made and sell where they're wanted. Every port has its own prices.

- **Fight.** Take prizes from your nation's enemies, hunt pirates, or turn pirate yourself.
- **Rise.** Win fame and favour, and climb from Gentleman Adventurer toward the peerage.
- **Meet the neighbours.** Mars, Venus, Mercury and the belt have peoples of their own. How you treat them decides how they treat you.
- **Stay alive.** Hull, provisions, coal and morale are your lifelines. Watch them all.

On the **title screen**, **Continue voyage** picks up where you left off and **New captain** starts afresh. In game, press **P** or **Esc** (or the Menu button) to pause, save, read this manual, or return to the title screen.

Aetherforge is in active development. Check this manual after each update; new systems arrive often.

Known Species

Seven peoples share the inner worlds with the colonists from Earth. Each keeps its own opinion of you, separate from the nations, and each is best at one officer's post.

	Species	Home	Best as
	**Ossari**	The canal cities of Mars. Ancient merchant princes who speak in chords.	Factor
	**Skarr**	The Martian high deserts. Proud raiders and duellists.	Marine captain
	**Lumari**	The Venusian swamps. Walking plant-folk, healers and naturalists.	Surgeon
	**Kesh**	The Venusian cloud highlands. Insect-folk who are never lost.	Navigator
	**Selenn**	Mercury's twilight belt. Crystal light-singers untouched by heat.	Lens-singer
	**Vorrn**	The Phaeton belt. Stout miners and engineers.	Engineer
	**Krillian**	Belt rocks and pirate havens. Salvagers who will buy anything.	Quartermaster
	**Humans**	Earth and every colony.	Any post, though not to the same height

What you trade, whom you attack and how you treat their officers all count with each people. Listen to them in the public houses; they know more than they say.

The Aether

Aetherforge's solar system works the way the Victorians believed it did. Every wonder here comes from a real theory of the day.

THE LUMINIFEROUS AETHER

Space is filled with a thin, invisible medium. Ether flyers sail it as ships sail the sea, driven by ether screws and lifted by apergy, a repulsive force drawn from rare crystals.

THE AETHER WIND

The Sun pushes the aether outward, so voyages away from the Sun run quickly and voyages toward it beat slowly against the current.

CALORIC

Heat is a fluid that soaks into hulls. Stray too near the Sun and your planking warps and your crew roasts.

THE WORLDS

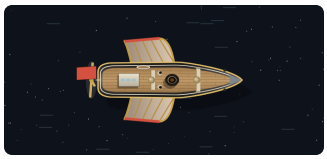
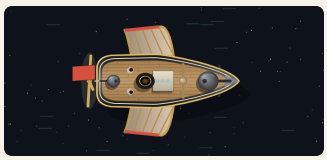
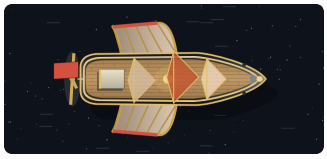
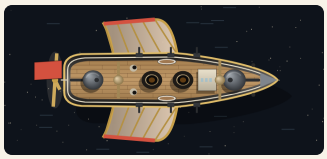
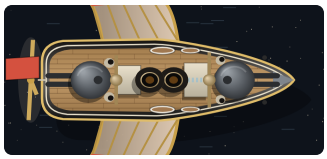
Mercury keeps one face to the Sun. Venus is a young world of swamps under endless cloud. Mars is old and drying, its cities clinging to their canals. Beyond Mars lies the Phaeton belt, the wreckage of a planet that once orbited there.

THE JOVIAN BARRIER

Past the belt the aether grows too thin and cold for any engine. No ship has crossed it.

Ship & Crew

SHIPS

	Ship	Notes
	**Packet sloop**	Small, nimble and cheap. Your first command.
	**Gunboat**	Heavy guns for her size, but poor against the aether wind.
	**Martian kite-galleon**	Sails well into the wind and carries plenty, but lightly built.
	**Armoured cruiser**	A strong all-rounder for a privateer.
	**Ironclad aether frigate**	Huge broadsides and a deep hold; slow to turn.

CREW

More hands mean faster reloads and stronger boarding parties, up to your ship's berths. Sign them on in public houses. Captured crews sometimes join you.

Provisions feed the crew: one load feeds twenty hands for a month. Run out and they'll hold out about a month before morale collapses into mutiny.

Morale slips slowly at sea, and rises when you make port, win prizes or stand the crew a round.

YOUR FLEET

A captured ship can join your fleet, up to eight ships in all, or become your new flagship. Only the flagship fights; the others follow in your wake, adding their holds and berths. The fleet sails at the pace of its slowest ship. Each ship needs a prize crew, so a fleet takes more hands, and if your crew falls too low the weakest ships are set adrift.

On the Fleet screen you can move your flag to another ship (names and upgrades stay with their ships), sell ships in port, or set them adrift at sea. A shipwright repairs the whole fleet at once. If your flagship is lost, your best remaining ship takes her place.

OFFICERS

Seven posts can each hold one officer: navigator, surgeon, engineer, factor, marine captain, quartermaster and lens-singer. Officers sign on for a small fee, draw a small monthly retainer, and take a share of every prize. They grow more skilled with time. They can be wounded or killed in battle, and they'll leave if they go unpaid, if morale collapses, if you go too long without prize money, or if you make enemies of their people.

Ports

MARKET

Each port makes some goods cheaply and pays well for others. Green prices are cheap for that port, amber are dear. Buying pushes the price up and selling pushes it down, and prices drift back over time.

SHIPWRIGHT

Repair your hull and vanes, or trade your ship in for another. The pictures show each ship to the same scale. Capital ports build the largest warships; Martian yards build kite-galleons.

Upgrades. Fit better ether vanes, guns, hull plating and boilers, each in four steps. Every yard has its specialities: Martian yards are masters of vanes, the Vorrn at Ceres of boilers, and the capitals of guns and armour. Hover over or press and hold an upgrade to see it on your ship. Upgrades stay with the ship, so they're left behind if you change ships, though they raise her trade-in value.

PUBLIC HOUSE

Sign on sailors, hire officers, and stand the crew a round. Listen to the patrons: rumours about rich ships and good prices are usually true.

GOVERNOR

Hear the news of wars, earn a letter of marque, and receive honours from your own nation. Martian cities have a palace instead.

A port where you're hated will refuse you docking.

Sailing

The planets move. Every world follows its true orbit, so the distance between Earth and Mars changes from month to month. A run between neighbouring worlds takes one to three weeks. Wait for a good alignment and you'll save days.

Read the compass. The blue arrow shows the aether wind; the brass needle shows your heading. Sailing with the wind is fastest. Some ships handle a sunward beat far better than others.

Boilers drive you at a steady speed whatever the wind, but burn coal. Keep some in the bunkers for emergencies, and to run from a fight.

Hazards. Aether squalls speed you along but batter the hull and vanes; steer out of them. The Sun's caloric blisters any ship that strays close. The Jovian Barrier marks the edge of the chart.

Heave to to stop and think. Time stands still while you're hove to.

Encounters

Sail close to a ship and she appears in the numbered list above your controls; press her number (or tap her) to approach. Hostile ships are marked.

When you approach a ship, through the telescope you'll see her flag, her type, roughly what she carries and whether she means trouble. You can **hail** her for news, **sheer off**, or **attack**.

Hostile ships may chase you and open fire. Fight, or stoke the boilers and run.

BATTLE

Your guns fire from the side facing the enemy, so turn to bring a broadside to bear. The two rings beside your ship show each broadside reloading.

- **Solid shot** reaches farthest and smashes the hull.
- **Chain shot** wrecks the ether vanes so she can't run.
- **Canister** is short-ranged and cuts down her crew.

A ship with no vanes or no fight left strikes her colours. Wreck her hull and she explodes. Break away far enough and the battle ends.

BOARDING DUELS



A chop, and the enemy captain ducks under it.

Lay your ship against hers and the grappling hooks fly. You may **let the crew settle it**, or **lead the boarders** and fight the enemy captain yourself.

In a duel every attack has one defence that stops it. The enemy's blow is called out as it is wound up, so watch for it.

- **Chop** is slow and drives the enemy back furthest. **Duck** under it.
- **Thrust** is quick and gains a little ground. **Parry** it.
- **Slash** lies between the two. **Jump** over it.

A blow that lands drives the loser back along the deck. Stop a blow and you may answer at once with a faster one. Drive the enemy captain to the far rail and the ship is yours; be driven to your own and you are beaten.

Meanwhile the crews fight behind you. The bar at the top shows who is winning; the side with the upper hand lends its captain speed. A **taunt** heartens your crew, but leaves you open. The sabre is balanced, the rapier quicker with a shorter guard, the cutlass slower but harder-hitting.

PRIZES

A captured ship gives up her gold, some of her crew and her cargo. You can add her to your fleet or make her your flagship. If you lose your flagship, your best remaining ship takes over; with no fleet, you'll be put ashore poorer but alive.

The Hunt

Somebody ruined your family, and you chose who at the start of the voyage. The hunt for them runs alongside everything else you do, at your own pace. **Captain**, then **The hunt**, opens a journal that always says what to do next.

FINDING THE LIEUTENANTS

Your nemesis works through three **lieutenants**, each with a ship. In any public house you can **buy a round and ask**. If nobody there knows, they will tell you which world to try; on the right world, somebody always talks.

Once you have a name, the lieutenant's ship is marked on the chart with a red ring, and the journal can **set a course** for her. They are hostile on sight, and each is stronger than the last, so build up your ship before you go looking.

THE PAPERS

Beat a lieutenant, by taking or sinking the ship, and you get the captain's **papers**. Each set tells you where one of your **family** is held, and names the next lieutenant.

FREEING YOUR FAMILY

Make port where one of them is held and a card appears at the top of the port screen. There are three ways:

- **Pay** the price asked. Each costs more than the last.
- **Appeal** to the governor or prince, who will listen only if you are respected there.
- **Break them out by night**: a sword fight against the captain of the guard, with your landing party at your back. Win or lose, that port will think the worse of you.

THE RECKONING

The last lieutenant's papers give away your nemesis's **stronghold**, which is then marked on the chart. Their flagship is an ironclad frigate with a full crew, and they are the best sword you will meet.

After the first lieutenant falls, your nemesis sends **hired hunters** after you from time to time. They show on the chart with a red ring.

Landings

Every world can be landed on. Sail close and choose **Land** from the list at the bottom of the chart, or tap the world itself. Days pass while you are ashore, and the crew still eats.



A landing party sets out along a dead Martian canal.

SITES AND SURVEYS

Each world has one to three **sites**. A site must be **surveyed** (two days) before you can use it. Every survey also earns a report, and the learned societies pay for reports when you next make port on Earth.

The Moon, the asteroids and Mercury's night side have no air. To set foot there you need **pressure suits**, sold once and for all at the shipwrights of the Earth capitals and at Ceres Haven.

DIGGING AND COLLECTING

- **Dig** where there is ore, coal, quicksilver, amber or apergy. A bigger crew and a good engineer dig more in a day. The easy diggings run out.
- **Collect specimens** where there is wildlife. The museums of Earth pay well for them, and now and then a collector doesn't come back.

Metal ore, **natural specimens** and **ancient relics** can't be bought anywhere. Earth pays best for all three.

EXPEDITIONS

Ruins and wild country are explored by a **landing party**. Choose how many hands go, how many days of water (or air) they carry, and which officers lead them. Supplies come out of your provisions.

The party advances one **stage** at a time, and each stage uses a day's supplies. At every stage something happens, and you choose what to do about it. Then you choose again: **push deeper** or **turn back**. The way home needs about half as many days as the way in. Run short and people die on the march out.

Officer ashore	What they do
Navigator	Finds safer paths and a shorter way home
Surgeon	Keeps the hurt alive
Chief engineer	Deals with bad ground and stuck doors
Marine captain	Leads the fighting
Lens-singer	Reads the old writing and finds more in ruins
Quartermaster	Packs out a bigger haul
Factor	Gets better terms from the peoples you meet

Officers who go ashore can be wounded. What you find at the deepest stage can only be taken once, but you can come back and go further another day.

OUTPOSTS AND COALING STATIONS



A mining outpost on an asteroid, with its palisade up.

On the Moon and the asteroids there is good ground to build on. First you need a **colonial charter**, sold by any great-power governor who thinks well of you. Then bring gold and **machinery** in your hold.

- A **coaling station** wins rock-coal from the ground, so you can fill your bunkers for nothing far from any port.
- A **mining outpost** works whatever the site holds: ore, quicksilver or apergy.

Both pile up cargo for you to load, keep a **warehouse** for your own goods, and can be enlarged twice. When the bunkers are full, the surplus is sold to passing ships at half price and the takings wait for you.

Pirates, and your charter's enemies in wartime, **raid** lonely stations. Leave hands as a **garrison** and build a **palisade and gun**. A beaten station loses its stores, and now and then is burned and must be rebuilt. The **Holdings** button on the chart lists your stations as last reported.

Nations

Britain, Germany, France and Russia each hold colonies on Mars and Venus, and the Martian city-states rule the rest. Wars start and end as months pass; the public houses and governors carry the news.

Attack a nation's ships and it turns against you. Attack its enemies and it warms to you. Pirates are fair game for everyone.

A **letter of marque** makes raiding an enemy's ships lawful and earns more fame. Fame and your own nation's favour bring **honours**, from Gentleman Adventurer upward, each with a reward.

The Martian courts have honours of their own for outlanders who earn their trust.

Network

The **heliograph network** lets you play alongside friends. Each captain keeps their own game, calendar and save; the network shows you each other and carries your signals.

OPENING AND JOINING A CIRCUIT

Open the pause menu and choose **Heliograph network**. One captain opens a circuit and gets a code, such as BRASS-417, to give to up to three friends. They choose Join a circuit and enter it. The circuit lasts while the captain who opened it keeps the game open.

MEETING IN THE AETHER

Captains on your circuit appear on your chart, ringed in brass and named. Because every voyage keeps its own calendar, a captain is shown by where she is relative to the nearest world: if you are both near Mars, you'll find each other near Mars. Sail close and she joins your numbered list; pick her to see her ship and send a private signal.

SIGNALS

From the Heliograph network screen you can signal every captain at once, or one privately, wherever they are.

Network play runs on the game's own website. Trading, sailing together and duels between captains come in later versions.

Saving

Your voyage saves itself whenever you make port and every ten days at sea. Use **Save voyage** in the pause menu to save at any moment.

Cloud backup. When you play on claude.ai, a private copy of your voyage is kept in your own account. If you clear your browser or move to another device, the title screen offers to restore it.

Snapshots. The last five saves are kept in the browser. Open Saves, Backups & Updates to go back to one.

Backup files. Save your voyage as a file to keep somewhere safe or to move between devices.

Updates. On claude.ai the game is always the newest version. A downloaded copy can check for updates and carries your voyage into the new file.

Controls

Where	Key	Action
Anywhere in game	P or Esc	Pause menu
The chart	Click or tap	Set a course to that point, port or ship
	Arrow keys, A or D	Steer by hand
	B	Stoke or bank the boilers
	Space	Heave to
	1 to 9	Pick from the numbered list of nearby ports, ships and landing places
	Enter	The first entry in that list
	+ and −, or mouse wheel	Zoom
Approaching a ship	Z	Attack
	X	Hail her
	C or Esc	Sheer off
Battle	Left and right arrows, A or D	Turn
	Up and down arrows, W or S	More or less sail (the top setting stokes the boilers)
	Z or ,	Fire solid shot
	X or .	Fire chain shot
	C or /	Fire canister
	B	Stoke or bank the boilers
	P or Esc	Pause
Expedition	1 to 9	Choose what to do
	Z or Enter	Push deeper
	X	Turn back
Boarding duel	Z or ,	Chop: slow, drives the enemy back furthest
	X or .	Thrust: quick, a short step back
	C or /	Slash: between the two
	Down arrow or S	Duck under a chop
	Left or right arrow, A or D	Parry a thrust
	Up arrow or W	Jump over a slash
	T	Taunt, to hearten your crew
On a phone	On-screen buttons	Battle has steering buttons on the left, and pause, sail, boilers and each kind of shot on the right

Aetherforge 0.34 · Beta · This manual describes the game as of version 0.34. Things change quickly, so check the in-game manual for the latest.